

Brief description of each file:

Plotsines.ipynb - basic design of plotting up to 10 sine waves simultaneously and the combined version of then. There are interactive sliders for omega, wave number, and amplitude. Operates in 1D across the x spatial domain and animations over time. Velocity syncing can be toggled where wave number and frequency are synced for each sine graph plotted based on the velocity.

2dmap.ipynb - 2d representation of sine waves (up to 10 waves) in 2D across the x and z spatial dimensions with a set height. Sliders for omega, wave numbers for x and z, and amplitude.

Triadic.ipynb - Has both 1D and 2D representations of sine graphs in different coding cells. Plots based on the formula $KX1 = KX2 + KX3$. Runs this triadic compatibility for creating different sine graphs in 1D or 2D.

Fielddata.ipynb - Maps out field data in 1D across x spatial dimension where $kz = 0$.

2Ddata_triadic_representation.ipynb - maps out field data in 2d across x and z spatial dimensions where $y = 86$ in triadic compatibility for omega, kx , and kz . There is an optional triadic compatibility component within the coding cell where random field data numbers are selected each time. Can have up to 25 triads for optimal time.

2Dfielddata.ipynb - maps out same field data without triadic compatibility limitation.